Year 10 Science
Physical Sciences
Worksheet - Equations of Motion (Extension)

DISPLACEMENT OF UNIFORMLY ACCELERATED BODIES

For any moving body: $v_{av} = \frac{s}{t}$

For a body moving with a uniform acceleration:

$$v_{av} = \frac{u + v}{2}$$

hence: $\frac{s}{t} = \frac{u + v}{2}$

$= \frac{u + (u + at)}{2}$ (as $v = u + at$)

$\therefore$ $\boxed{s = ut + \frac{1}{2} at^2}$

TYPE EXAMPLE 13: A vehicle travelling at 10 m s^{-1} is accelerated at 4 m s^{-2}. After a period of 5 s, determine the:
(a) displacement
(b) average velocity
(c) final velocity.

(a) $u = 10$ m s^{-1}
$a = 4$ m s^{-2}
$t = 5$ s
$s = ?$

$s = ut + \frac{1}{2} at^2$
$= (10 \times 5) + \frac{1}{2} (4 \times 5^2)$
$= 50 + \frac{1}{2} (4 \times 25)$
$= 50 + 50$
$= 100$ m

i.e. the displacement is 100 m (in the direction of the initial motion).

(b) $v_{av} = ?$
$s = 100$ m
$t = 5$ s

$v_{av} = \frac{s}{t}$
$= \frac{100}{5}$
$= 20$ m s^{-1}

i.e. the average velocity is 20 m s^{-1} (in the direction of the initial motion).

(c) $v = ?$
$u = 10$ m s^{-1}
$a = 4$ m s^{-2}
$t = 5$ s

$v = u + at$
$= 10 + (4 \times 5)$
$= 10 + 20$
$= 30$ m s^{-1}

i.e. the final velocity is 30 m s^{-1} (in the direction of the initial motion).

OR $v = ?$
$u = 10\ m\ s^{-1}$
$v_{av} = 20\ m\ s^{-1}$

$$v_{av} = \frac{u + v}{2}$$
$$\therefore u + v = 2 \times v_{av}$$
$$\therefore v = (2 \times v_{av}) - u$$
$$= (2 \times 20) - 10$$
$$= 40 - 10$$
$$= 30\ m\ s^{-1}$$

i.e. the final velocity is 30 m s^{-1} (in the direction of the initial motion).

SET 10: DISPLACEMENT OF UNIFORMLY ACCELERATED BODIES I

1. A car moving with a velocity of 20 m s^{-1} is accelerated at 3 m s^{-2}. After 8 s, determine the:
   (a) displacement.
   (b) average velocity.
   (c) final velocity.
2. What distance will a cyclist travel if when starting from rest the cycle is accelerated at 1.8 m s^{-2} for 4 s?
3. A motor boat moving with a velocity of 2 m s^{-1} is accelerated at 1 m s^{-2} for 6 s. Calculate the:
   (a) displacement.
   (b) final velocity.
   (c) average velocity.
4. A train, initially at rest, leaves a railway station with an acceleration of 0.4 m s^{-2} for a period of 50 s. Find the:
   (a) displacement.
   (b) final velocity.
   (c) average velocity.
5. A truck travelling at 15 m s^{-1} brakes uniformly at 1.75 m s^{-2} for 4 s in coming to rest at a stop sign. At what distance from the sign did the truck commence braking?
6. A ball starting from rest rolls down an inclined plank with an average acceleration of 5 m s^{-2}. If it takes 2 s to roll off the end, determine the:
   (a) length of the plank.
   (b) final velocity of the ball.
   (c) average velocity of the ball.
7. A bicycle moving at 8 m s^{-1} is accelerated at -1.25 m s^{-2} for a 6 s period. Determine the:
   (a) displacement.
   (b) velocity at the end of this period.
8. A body with a speed of 50 m s^{-1} is subjected to a uniform retardation of 10 m s^{-2} for 4 s. Determine the:
   (a) distance travelled.
   (b) average speed.
   (c) final speed.
9. A plane can effect take-off with a velocity of 72 m s^{-1}. If the plane can accelerate along the runway at 4 m s^{-2}, determine the:
   (a) time taken to reach a speed of 72 m s^{-1}
   (b) minimum length of runway to enable a safe take-off.
10. A person starting from rest takes 6 s to reach a maximum velocity down a ski slope. Assuming a constant acceleration of 5 m s^{-2} find the:
    (a) distance travelled in 6 s
    (b) maximum velocity attained by the skier.

SET 10

1. (a) 256 m (b) 32 m s^{-1} (c) 44 m s^{-1} **2.** 14.4 m
3. (a) 30 m (b) 8 m s^{-1} (c) 5 m s^{-1} **4.** (a) 500 m (b) 20 m s^{-1} (c) 10 m s^{-1}
5. 46 m **6.** (a) 10 m (b) 10 m s^{-1} (c) 5 m s^{-1} **7.** (a) 25.5 m (b) 0.5 m s^{-1}
8. (a) 120 m (b) 30 m s^{-1} (c) 10 m s^{-1} **9.** (a) 18 s (b) 648 m
10. (a) 90 m (b) 30 m s^{-1}

ANOTHER EQUATION OF MOTION

The equation, $v^2 = u^2 + 2as$ is often used to solve in one step problems which require the use of both of the other two equations of motion, $v = u + at$, and $s = ut + \frac{1}{2} at^2$.

GRAVITATIONAL ACCELERATION

All freely falling bodies close to the earth's surface are accelerated at the same rate (i.e. they fall the same distance in the same time). The acceleration due to gravity is often referred to as g.

Acceleration due to gravity acts in a vertically downward direction and has a value of approximately 10 m s^{-2}.

The previously established equations of motion for uniformly accelerated bodies may be used for freely falling bodies.

$$v = u + gt$$
$$s = ut + \frac{1}{2} gt^2$$

where: g = acceleration due to gravity
$= 10$ m s^{-2} (vertically downward)

NOTE: 1. The effects of frictional air resistance are ignored and are assumed to be negligible.

2. At ground level g has a value of about 9.8 m s^{-2}. For ease of calculation the rounded off value of 10 m s^{-2} is used.

TYPE EXAMPLE 14: A stone falls off the edge of a cliff. If it takes 6 s to strike the ground below, determine the:

(a) velocity with which the stone strikes the ground.

(b) height of the cliff.

(a) $u = 0$, $t = 6\ s$, $g = 10\ m\ s^{-2}$, $v = ?$

$v = u + gt$
$= 0 + (10 \times 6)$
$= 60\ m\ s^{-1}$

i.e. the stone strikes the ground at $60\ m\ s^{-1}$ (downwards).

(b) $u = 0$, $t = 6\ s$, $g = 10\ m\ s^{-2}$, $s = ?$

$s = ut + \frac{1}{2} gt^2$
$= 0 + \frac{1}{2} (10 \times 6^2)$
$= \frac{1}{2} (10 \times 36)$
$= 180\ m$

i.e. as the displacement is 180 m (downwards) the height of the cliff is 180 m.

SET 12: GRAVITATIONAL ACCELERATION I

NOTE: Use acceleration due to gravity, $g = 10\ m\ s^{-2}$.

1. A stone drops freely from rest. Determine after 4 s the:
   (a) stone's final velocity
   (b) displacement.
2. A can of paint falls from the top of a painter's ladder. If it takes 1.2 s for the can to strike the ground below, find the:
   (a) final velocity of the can
   (b) length of the ladder (assumed to be vertical).
3. A cricket ball is hit vertically into the air. The ball at its maximum height is momentarily at rest and from this position takes 2.5 s to strike the ground. Calculate the height to which the ball was hit.
4. A stone is dropped from rest into a well. Find the depth of the well if it takes 0.9 s for the stone to strike the bottom.
5. A ball is thrown vertically downwards at $10\ m\ s^{-1}$. How far does the ball travel in 3 s?
6. An object drops (from rest) out of a window of a tall building. How far does the object fall:
   (a) in 2 s?
   (b) in 3 s?
   (c) in the 3rd second?

SET 12

1. (a) $40\ m\ s^{-1}$ (b) 80 m **2.** (a) $12\ m\ s^{-1}$ (b) 7.2 m **3.** 31.25 m
4. 4.05 m **5.** 75 m **6.** (a) 20 m (b) 45 m (c) 25 m

ACCELERATION DUE TO GRAVITY (EXTENDED)

When a projectile has an initial velocity upwards the sign convention *MUST* be used as 'g' is *ALWAYS* directed vertically downards. As it rises the projectile slows until at its maximum height its velocity is *MOMENTARILY* zero. This occurs because the initial velocity and acceleration vectors are in opposite directions. From its maximum height the projectile falls with a velocity increasing in the downward direction until, at the moment it returns to the position from which it was projected, it is of the same magnitude as when it was projected vertically but in the *OPPOSITE* direction. This occurs because the final velocity and acceleration vectors are in the same direction.

For the entire journey g is constant (and equal to 10 m s^{-2} downwards).

Generally downward vectors are designated a *POSITIVE* sign.

TYPE EXAMPLE 15: A stone is thrown vertically upward with a velocity of 35 m s^{-1}. Find the velocity of the stone after

(a) 2 s

(b) 4 s.

(a)

$g = +10\ m\ s^{-2}$
$u = -35\ m\ s^{-1}$
$t = 2\ s$
$v = ?$

$v = u + gt$
$= -35 + (10 \times 2)$
$= -35 + 20$
$= -15\ m\ s^{-1}$

i.e. the velocity after 2 s is 15 m s^{-1} UPWARDS.

(b)

$g = +10\ m\ s^{-2}$
$u = -35\ m\ s^{-1}$
$t = 4\ s$
$v = ?$

$v = u + gt$
$= -35 + (10 \times 4)$
$= -35 + 40$
$= 5\ m\ s^{-1}$

i.e. the velocity after 4 s is 5 m s^{-1} DOWNWARDS.

To determine the maximum height reached by a projectile during its flight the following may assist:

1. *at maximum height, the velocity is momentarily zero.*
2. *the time taken to reach maximum height is the same as the time taken to return from the position of maximum height.*

TYPE EXAMPLE 16: An arrow is fired vertically upwards with a velocity of 80 m s^{-1}.

Determine the: (a) time taken to reach maximum height.
(b) maximum height reached by the arrow.

(a) $g = +10 \text{ m s}^{-2}$, $u = -80 \text{ m s}^{-1}$, $v = 0$, $t = ?$

$$v = u + gt$$
$$\therefore gt = v - u$$
$$\therefore t = \frac{v - u}{g}$$
$$= \frac{0 - (-80)}{10}$$
$$= 8 \text{ s}$$

i.e. the time taken to reach maximum height is 8 s.

(b) *Consider the arrow falling from its position of maximum height.*

The time taken to reach maximum height is equal to the time taken for the arrow to return $\therefore t = 8$ s.

$g = 10 \text{ m s}^{-2}$, $t = 8$ s, $u = 0$, $s = ?$

$$s = ut + \tfrac{1}{2} gt^2$$
$$= 0 + \tfrac{1}{2} (10 \times 8^2)$$
$$= \tfrac{1}{2} (10 \times 64)$$
$$= 320 \text{ m}$$

i.e. As the displacement of the falling arrow is 320 m (DOWNWARDS) the maximum height reached by the arrow is 320 m.

OR

Consider the arrow rising to its position of maximum height.

$s = ?$, $g = 10 \text{ m s}^{-2}$, $u = -80 \text{ m s}^{-1}$, $t = 8$ s (from part a)

$$s = ut + \tfrac{1}{2} gt^2$$
$$= (-80) \times 8 + \tfrac{1}{2} (10 \times 8^2)$$
$$= -640 + 320$$
$$= -320 \text{ m}$$

i.e. As the displacement is 320 m (UPWARDS) the maximum height reached by the arrow is 320 m.

SET 13: GRAVITATIONAL ACCELERATION II

NOTE: Use acceleration due to gravity, $g = 10\ m\ s^{-2}$.

1. A stone is thrown vertically downwards at 10 m s^{-1}. After 2.5 s, find the:
   (a) velocity.
   (b) displacement.
2. A ball is thrown vertically downwards into a well with a velocity of 15 m s^{-1}. If the time taken to strike the water is 1 s find the:
   (a) velocity with which the stone strikes the water.
   (b) depth of the well to the water level.
3. A bullet is fired vertically upwards at 600 m s^{-1}.
   What is the velocity of the bullet after:
   (a) 20 s?
   (b) 100 s?
4. A ball is thrown vertically upwards at 13 m s^{-1}. What is its velocity after 1.6 s?
5. A bullet is fired vertically upwards at 600 m s^{-1}. Determine the:
   (a) time taken for the bullet to reach the maximum height
   (b) height reached by the bullet.
6. A stone is thrown vertically upward at 16 m s^{-1}. Calculate the:
   (a) time taken to reach maximum height
   (b) height reached by the stone.
7. A brick falls from rest off the top of a 45 m tall construction site. How long does it take to strike the ground?
8. How long would it take for a parcel to strike the ground if dropped from rest out of a helicopter stationary at a height of 80 m?
9. A stone is dropped from rest. Calculate the:
   (a) time taken to fall 12.8 m
   (b) velocity at this displacement.
10. An object is thrown vertically upward at 17.2 m s^{-1}. How long will it take for the object to return to the ground?

SET 13

1. (a) 35 m s^{-1} (b) 56.25 m **2.** (a) 25 m s^{-1} (b) 20 m
3. (a) 400 m s^{-1} upwards (b) 400 m s^{-1} downwards
4. 3 m s^{-1} downwards **5.** (a) 60 s (b) 18 000 m **6.** (a) 1.6 s (b) 12.8 m
7. 3 s **8.** 4 s **9.** (a) 1.6 s (b) 16 m s^{-1} **10.** 3.4 s